VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 5 – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. A 9 Hz wave is sampled at 12 Hz. Which frequency do the samples appear to have? What is the minimum alias-free sampling rate? (5 pts)

1.2. When should you prefer **RANSAC**, and when the **Hough transform**? Give one distinctive advantage of each. (5 pts)

1.3. Write the reflection across the horizontal (x) axis and its determinant. What does a negative determinant mean? Composing the reflections across both axes gives which transformation? (5 pts)

1.4. Two images of the same scene and viewpoint, but the second is brighter and has higher contrast. Which distance should compare them? Explain. (5 pts)

Problem 2. Image Filtering

2.1. A 7×7 image has a 2-pixel-thick **horizontal** line in rows 3–4 (value 1, else 0). Convolve **vertically** with $[-1, 0, 1]^\top$ (zero padding). (a) Write one column of the result. (b) How many local maxima (along columns)? How many for |response|? (c) How does the result change with correlation instead of convolution? (13 pts)

2.2. (a) By the $2\lceil 3\sigma \rceil + 1$ rule, what kernel size does $\sigma = 1.4$ need? Why renormalize the sum to 1 after truncation? (b) Blurring a uniform white image (255) with a Gaussian and zero padding: what happens at the border? Which border mode fixes it? (12 pts)

Problem 3. Feature Detection and Matching

3.1. After NMS, an 8-connected chain of pixels along a curve has (normalized) magnitudes $[0.9, 0.5, 0.45, 0.5, 0.2, 0.6, 0.95]$. Canny thresholds: low $= 0.4$, high $= 0.8$. (a) Classify each pixel. (b) Which pixels are edges after hysteresis? (c) What would a single threshold of 0.8 give? Of 0.4 over the whole image? (12 pts)

3.2. SIFT keypoint localization (1-D in x for simplicity): $D(-1) = 0.2$, $D(0) = 0.5$, $D(1) = 0.4$ (image in $[0, 1]$). (a) With central differences, find $\hat{x}$ and $D(\hat{x})$. Keep or reject with the 0.03 threshold? (b) Spatial Hessian $\begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$: does it pass the edge test with $r = 10$? (13 pts)

Problem 4. Fitting and Image Transformations

4.1. Homography estimation with the DLT. (a) Write the two rows of A for the pair $(2, 3) \to (5, 7)$. (b) Why are at least 4 pairs, no three collinear, required? (c) Why normalize coordinates (Hartley) before solving with the SVD? (12 pts)

4.2. Vector $\mathbf{x} = (5, -2, 4)$. A new coordinate system is obtained from the original by translating

by $(1, 2, 3)$, followed by a rotation about the Z axis by $-90°$. What are $\mathbf{x}$'s coordinates in the new system? Write the 4×4 homogeneous matrix performing this change of coordinates. (13 pts)

Problem 5. System Design

5. A fixed traffic camera looks obliquely down a straight road. Design a system that **estimates each vehicle's speed** (km/h) from video alone, given only the real-world size of a few road markings. (5 pts)